LAB # 10
Explain and Reproduce the Working of 4×3 Matrix Keypad with STM32F407 Using HAL Library

Objectives

- Understand the operational principles and scanning techniques of matrix keypads
- Configure GPIO pins using STM32 HAL library for keypad interfacing
- Display keypad input on 7-segment displays and LCD modules

Pre-Lab Exercise

Read the details given below in order to comprehend the basic operation of Keypad, and interfacing of keypad with microcontroller. Keypad can also be interfaced with LCD or seven segment to perform different tasks in different combinations.

What is a Matrix Keypad?

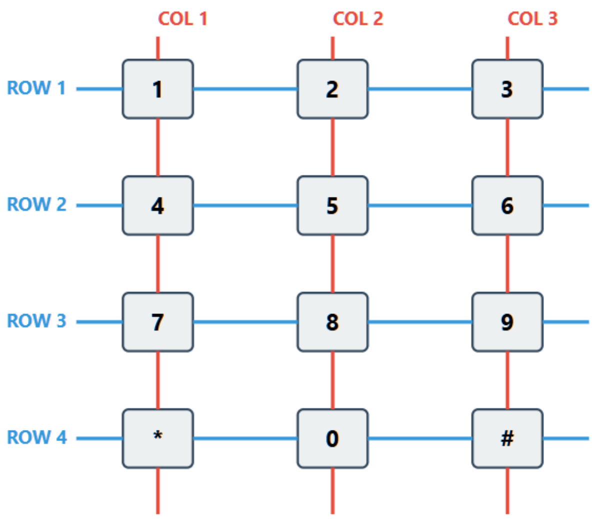
A matrix keypad is an intelligent input device that allows users to enter numeric and symbolic data efficiently. Unlike individual push buttons that require one GPIO pin per button, matrix keypads use a **row-column arrangement** that dramatically reduces the number of pins needed.

Why Matrix Configuration?

Consider a 4×3 keypad with 12 keys:

- **Individual wiring**: Would require 12 GPIO pins
- **Matrix arrangement**: Requires only 7 GPIO pins (4 rows + 3 columns)
- **Pin savings**: 42% reduction in GPIO usage!

This efficient architecture makes matrix keypads ideal for embedded systems where GPIO pins are precious resources.



How Keypad Scanning Works

The scanning process follows these steps:

1. Row Activation Phase

- Set one row to logic LOW (0V)
- Keep all other rows HIGH (3.3V)
- Configure rows as outputs

2. Column Reading Phase

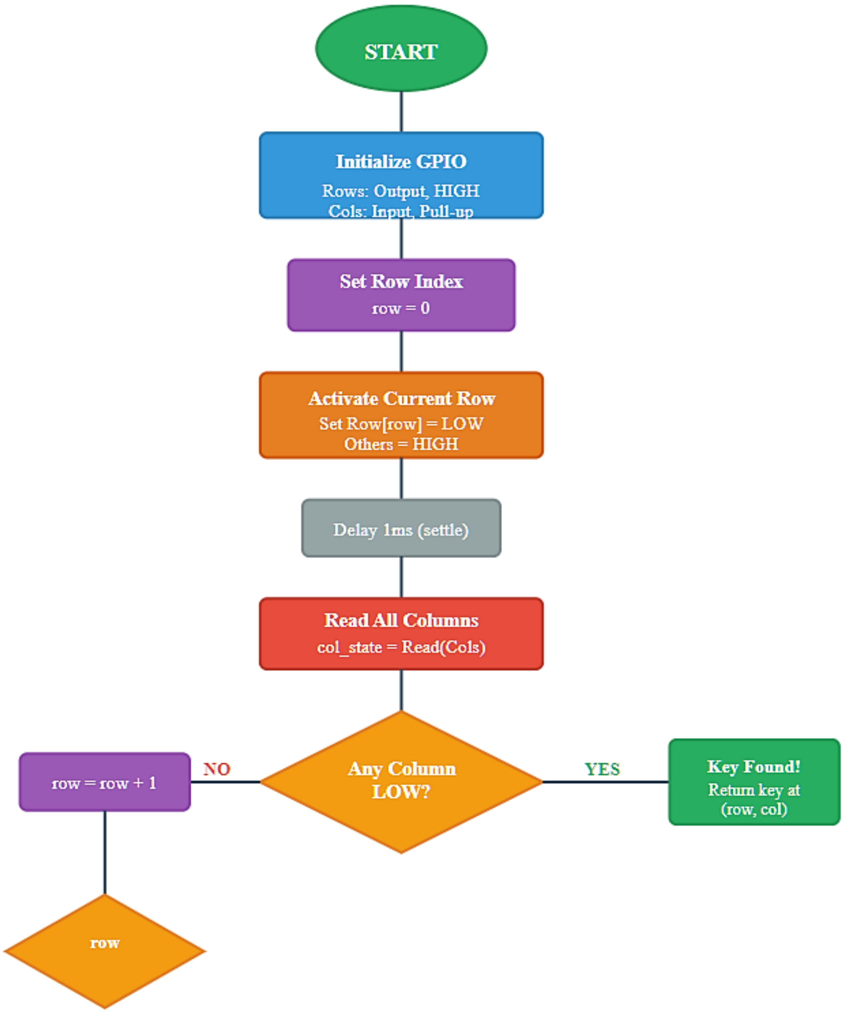
- Configure columns as inputs with pull-up resistors
- Read the state of all columns
- A LOW reading on a column indicates a key press at the intersection

3. Key Detection Logic

```
IF (Row_N is LOW) AND (Column_M reads LOW)
THEN Key at position (N,M) is pressed
```

4. Multiplexing Cycle

- Repeat for each row sequentially
- Complete scan cycle: typically 10-20ms
- Fast enough for real-time human interaction



STM32F407 GPIO Configuration Using HAL Library

Pin Assignment for Trainer Board

The STM32F407 trainer board connects the 4×3 keypad to **Port D**:

Function	Pin	Configuration	Voltage Level
ROW 1	PD7	Output (Push-Pull)	Active LOW
ROW 2	PD6	Output (Push-Pull)	Active LOW
ROW 3	PD4	Output (Push-Pull)	Active LOW
ROW 4	PD2	Output (Push-Pull)	Active LOW
COL 1	PD1	Input (Pull-Up)	Read HIGH/LOW
COL 2	PD3	Input (Pull-Up)	Read HIGH/LOW
COL 3	PB3	Input (Pull-Up)	Read HIGH/LOW

In-Lab Exercise

Task 1: Keypad Scanning

Complete the program that continuously scans the keypad and display the value of pressed key on 7-Segment (Lab8).

```
char Keypad_ScanRow1(void)
{
    /* Step 1: Activate Row 1 by setting it LOW */
 HAL_GPIO_WritePin(GPIOD, GPIO_PIN_7, GPIO_PIN_RESET);

    /* Step 2: Check Column 1 (Key '1') */
    if (HAL_GPIO_ReadPin(GPIOD, GPIO_PIN_1) == GPIO_PIN_RESET)
    {
        /* Column 1 is LOW - Key '1' is pressed */
        HAL_GPIO_WritePin(GPIOD, GPIO_PIN_7, GPIO_PIN_SET);  // Deactivate row
        return '1';
    }

    /* Step 3: Check Column 2 (Key '2') */
    if (HAL_GPIO_ReadPin(GPIOD, GPIO_PIN_3) == GPIO_PIN_RESET)
    {
        /* Column 2 is LOW - Key '2' is pressed */
        HAL_GPIO_WritePin(GPIOD, GPIO_PIN_7, GPIO_PIN_SET);  // Deactivate row
        return '2';
    }

    /* Step 4: Check Column 3 (Key '3') */
    if (HAL_GPIO_ReadPin(GPIOB, GPIO_PIN_3) == GPIO_PIN_RESET)
    {
        /* Column 3 is LOW - Key '3' is pressed */
        HAL_GPIO_WritePin(GPIOD, GPIO_PIN_7, GPIO_PIN_SET);  // Deactivate row
        return '3';
    }

    /* Step 5: No key pressed - Deactivate Row 1 */
    HAL_GPIO_WritePin(GPIOD, GPIO_PIN_7, GPIO_PIN_SET);
    return '\0';  // No key pressed
}
```

```
int main(void)
{
  HAL_Init();
  SystemClock_Config();
  MX_GPIO_Init();
        char key;

            while (1)
            {
                /* Scan keypad */
              key = Keypad_ScanRow1();

       if (key != '\0')
         {

                    /* Valid key pressed */
         }
                HAL_Delay(10);  // Scan every 10ms for responsive feel
            }

    }
```

Task 2: Write a complete C program to interface a Keypad and LCD with STM32F4

Write a complete C program that will interface a keypad with STM32F4 microcontroller and continuously scans it. Whenever a key is pressed, the corresponding character will be displayed on the LCD. Verify your program by running it on the STM32F4 Discovery board.

: Scrolls from left to right on the upper line

- **First Line:** 4x3 Keypad
- **Second Line:** The input is :

Rubric for Lab Assessment

The student performance for the assigned task during the lab session was:			
Excellent	The student completed all assigned tasks independently, strictly followed all health and safety protocols, and presented the results clearly and accurately.	4	
Good	The student completed the assigned tasks with minimal guidance, mostly followed health and safety protocols, and presented the results appropriately.	3	
Average	The student partially completed the assigned tasks, inconsistently followed health and safety protocols, and presented partial or unclear results.	2	
Worst	The student did not complete the assigned tasks, frequently disregarded health and safety protocols, and failed to present results.	1	

Instructor Signature: ____________________________ Date: _________________